MILITARY SPECIFICATION SHEET

INTERCOMMUNICATIONS SETS, AN/AIC-14 and AN/AIC-14A

AMPLIFIER MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope. This specification sheet provides performance and physical characteristics for the amplifier used in two places in master Interphone Control panel C-2645D/AIC-14A as (1) an interphone amplifier and as (2) an isolation amplifier.

1.2 Military Part Number. The military part number shall be M22352/3-1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

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3. REQUIREMENTS

3.1 Item Definition. The Amplifier Module shall provide amplification of audio signals with gain characteristics as specified herein when used in Master Interphone Control panel C-2645D/AIC-14A.

3.1.1 Item Diagram. The Amplifier Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface Definition. The Amplifier Module input and output signals will normally display the characteristics presented in Table I.

TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
No. 1 (NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
No. 2 (NO. 2)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Power (PWR)	+27.5 ± 0.5 Vdc
Outputs:	
Headset High (HDST HI Z)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 19.4 Vac.
Headset Low (HDST LO Z)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 2.2 Vac.
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

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3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 No. 1 (NO. 1). The Amplifier Module shall have a NO. 1 audio input circuit having the following performance characteristics:

3.2.1.1.1.1 Signal Frequency. The amplifier shall accept input drive signals which range from 300 to 7,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The amplifier shall accept input drive signals which range in amplitude from 0.01 to 0.10 Vac. The amplifier shall be capable of withstanding overvoltage inputs of up to 1.0 Vac without sustaining damage. The amplifier shall be capable of operating within specification when the signal input contains a dc component of up to 2 volts.

3.2.1.1.1.3 Input Impedance. The amplifier shall input impedance of 1200 ohms ±10 percent.

3.2.1.1.1.4 Isolation. The isolation between signal inputs shall be not less than 20 dB.

3.2.1.1.2 No. 2 (NO. 2). The NO. 2 input characteristic shall be as described in Section 3.2.1.1.1 for Input NO. 1.

3.2.1.1.3 Power (PWR). The Amplifier Module shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the amplifier shall not exceed 200 milliamperes.

3.2.1.2 Output Characteristics. The Amplifier Module shall have the following output characteristics:

3.2.1.2.1 Headset High (HDST HI Z). The HDST HI Z output shall have the following characteristics:

3.2.1.2.1.1 Frequency and Amplitude. The HDST HI Z signal output shall exceed the input amplitude(0.06 Vac) at 1,000 Hz by 44 ±1dB (8.5 to 10.7 Vac) when terminated in 600 ohms. The HDST HI output signal amplitude shall be in accordance with the response curves of Figure 2 when terminated in a 600 ohm resistive load. The input/output characteristic (gain) of the amplifier shall be linear within ±2dB when the input level is varied between 0.01 and 0.10 volts rms.

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3.2.1.2.1.2 Power The amplifier shall be capable of producing an output of 500 milliwatts into a non-inductive load of 270 to 600 ohms at 1,000 Hz. The amplifier shall be capable of producing an output of not less than 250 milliwatts into a non-inductive load of 601 to 1,000 ohms.

3.2.1.2.1.3 Output Impedance. The HDST HI Z output shall have a nominal impedance of 600 ohms ±10 percent.

3.2.1.2.1.4 Distortion The total harmonic distortion over the frequency range of 300 to 7,000 Hz and input amplitude range of 0.01 to 0.10V shall not exceed 5 percent under standard conditions as defined in Section 3.1a of MIL-STD-810. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 8 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.2.1.2.1.5 Phasing The output shall lag the input by approximately 180 degrees at 1,000 Hz.

3.2.1.2.1.6 Amplifier Stability. The amplifier which produces the HDST HI Z output signal shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or any combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

3.2.1.2.1.7 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated throughout the environmental envelope as defined in MIL-E-5400 for Class 2 equipment with the input circuit shorted to system ground.

3.2.1.2.1.8 Susceptibility to Stray Magnetic Fields. The amplifier shall be shielded or so designed that when terminated into a 600 ohm load within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz irrespective of orientation.

3.2.1.2.2 Headset Low (HDST LO Z). The HDST LO Z output shall have characteristics identical to those described in 3.2.1.2.1 except that the output impedance shall be 7.5 ohms ±10 percent at 1000 Hz and the output voltage amplitude correspondingly 0.95 to 1.2 Vac for the same input voltage (0.06 Vac). Output loads for frequency response (see Figure 2), power, amplification, and distortion tests shall be 7.5 ohms ±10 percent.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The Amplifier Module shall consist of a single encapsulated module having the dimensions of Figure 3. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The amplifier shall weigh no more than 4.5 ounces.

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3.3 Design and Construction. The design and construction of the amplifier shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 of MIL-I-22352(AS).

4.3 Test Methods.

4.3.1 Examination of product See 4.3.3.2.1 of MIL-I-22352(AS).

4.3.2 Operational test. See 4.3.3.2.2 of MIL-I-22352(AS).

4.3.2.1 Input Impedance. Verify that the NO. 1 and NO. 2 input impedances are in accordance with 3.2.1.1.1.3 herein.

4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.2.1.2.1.1 herein.

4.3.2.3 Frequency Response. Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements of 3.2.1.2.1.1 herein.

4.3.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and 0.10 Vac, verify that the amplifier meets the requirements of 3.2.1.2.1.1.

4.3.2.6 Distortion. Verify the requirements of 3.2.1.2.1.4 herein.

4.3.2.7 Phasing. Verify the requirements of 3.2.1.2.1.5 herein.

4.3.2.8 Amplifier stability. Verify the requirements of 3.2.1.2.1.6 herein.

4.3.2.9 Internal noise. Verify the requirements of 3.2.1.2.1.7 herein.

4.3.2.10 Susceptibility to stray magnetic fields. Verify the performance of 3.2.1.2.1.8 herein.

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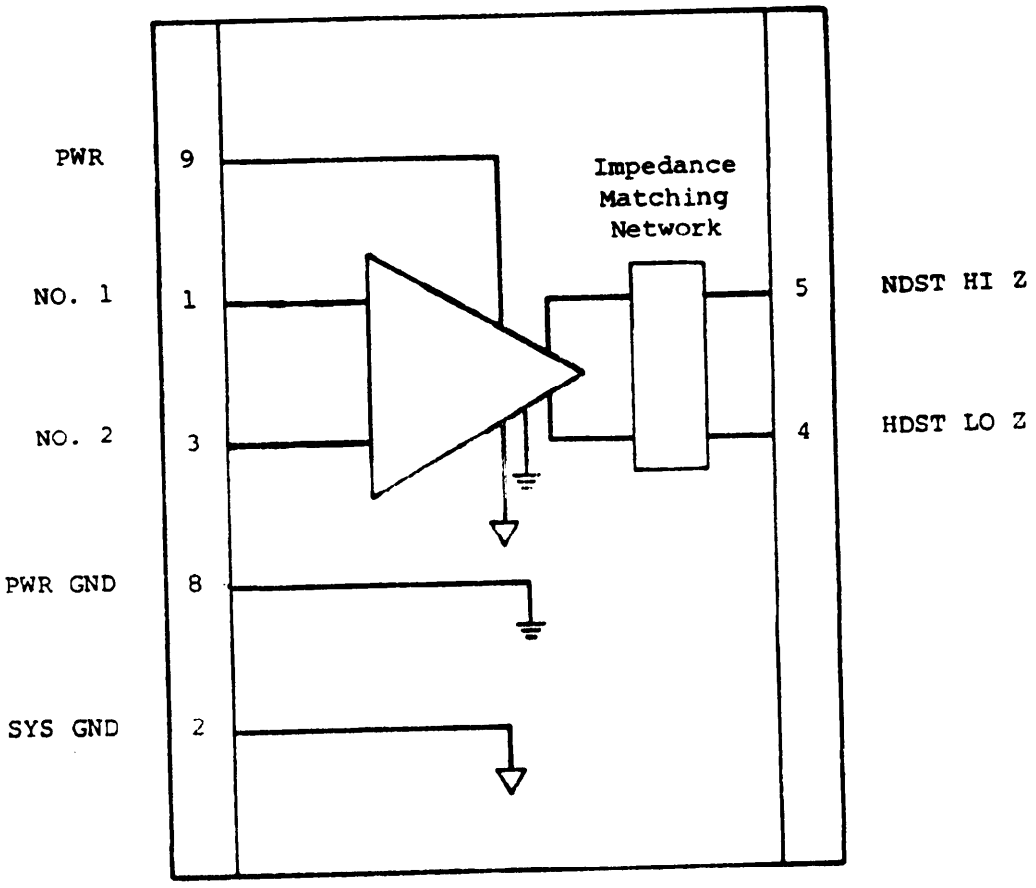
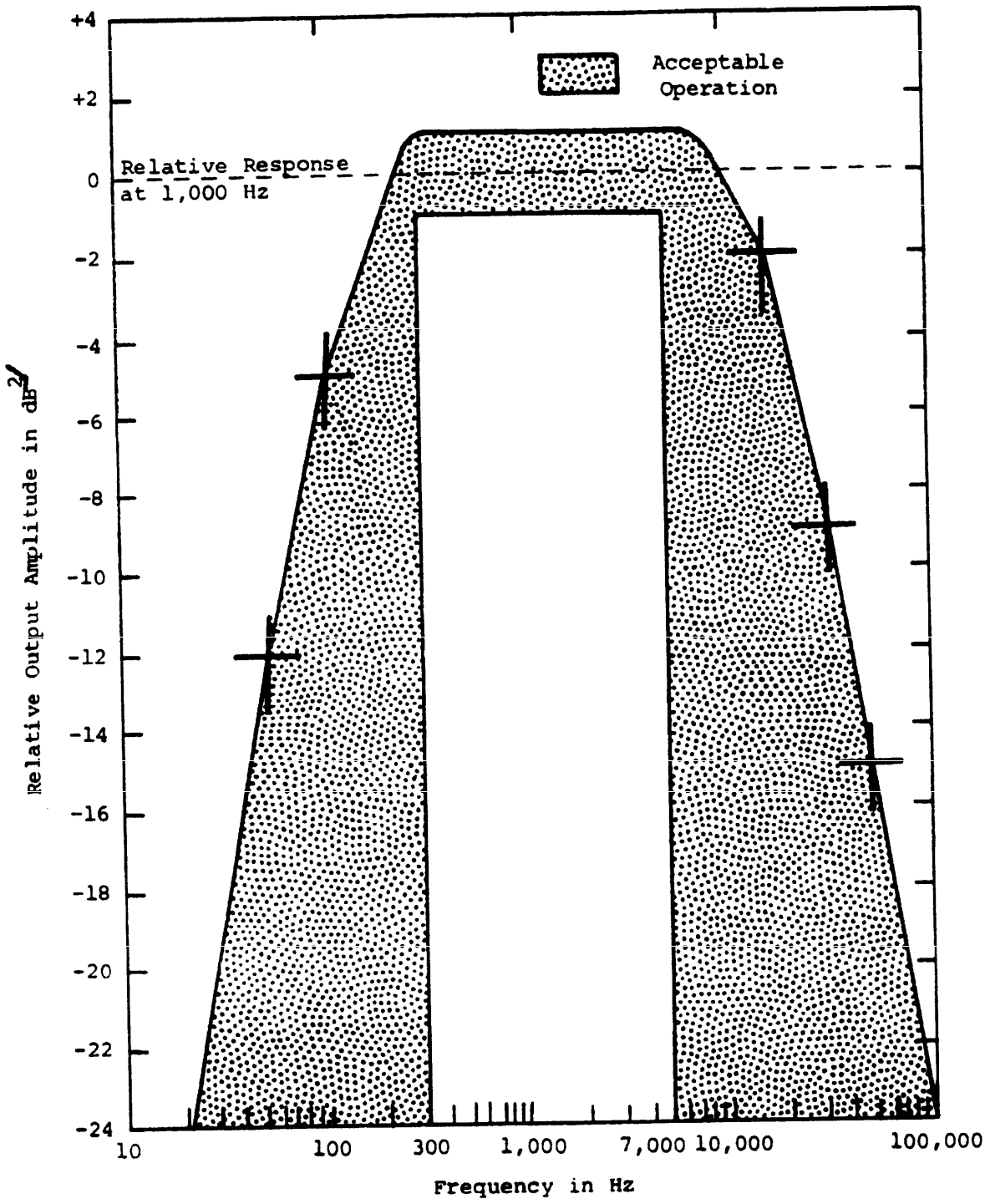


FIGURE 1. Amplifier module functional diagram.

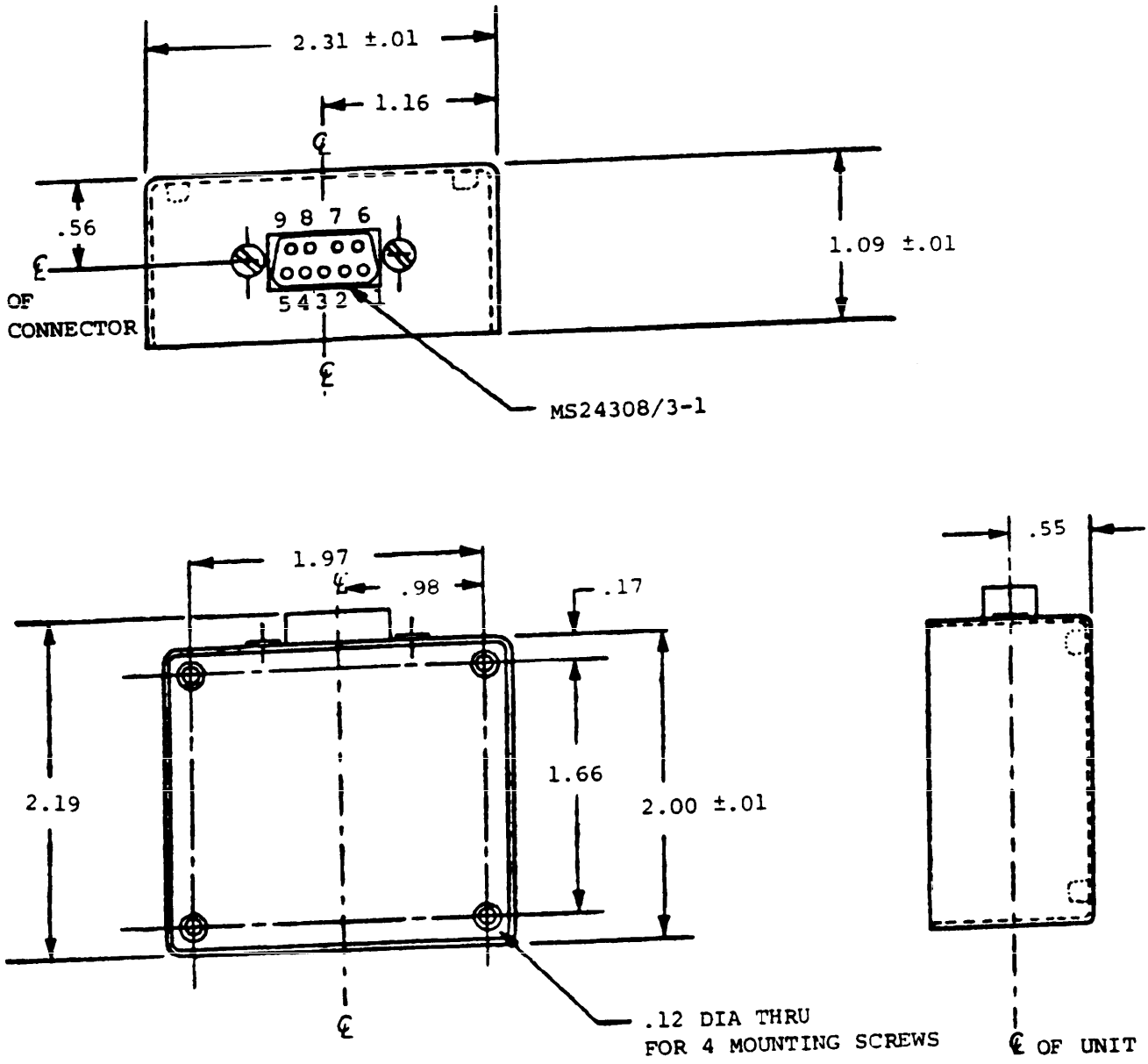
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1/ Termination into 600 Ohms.
2/ Output amplitude is measured with respect to the amplitude at 1,000 Hz.

FIGURE 2. Amplifier module frequency response.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 3. Amplifier module outline drawing.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/3(AS)	INTERCOMMUNICATIONS SETS, AN/AIC-14 AND AN/AIC-14P AMPLIFIER MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

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